RESEARCH OVERVIEW & PROJECT INTRODUCTION

A MACHINE LEARNING FRAMEWORK FOR DAILY RAINFALL CLASSIFICATION & EXTREME EVENT ANALYSIS

10-Year Multi-Source Ground-Satellite Synthesis (2016–2025) in the Kathmandu Watershed, Nepal

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10 Years
Temporal Coverage
2016–2025 Complete Daily Data

19 Stations
Terrestrial Network
Screened DHM Rain Gauges

69,407
Observation Records
Station-Day Ground Points

~650 km²
Watershed Domain
1,280 m to 2,150+ masl Relief

5 Classes
Intensity Schemes
Standard WMO / DHM Tiers

0.1° × 0.1°
Satellite Ingestion
NASA GPM IMERG V07 Gridded

RESEARCH BACKGROUND, PROBLEM CONTEXT & OBJECTIVES

MOTIVATION & HAZARD VULNERABILITY

- Steep Himalayan Orography & Rapid Urbanization: Kathmandu Valley hosts >2.5 million residents in a complex, bowl-shaped basin.
- Severe Monsoon Flood Risk: >80% of precipitation falls during June–September; constrained river corridors cause catastrophic flash floods.
- Climate Hazard Vulnerability: Devastating cloudburst disasters (e.g. September 2024 flood) highlight the critical need for localized warnings.

CORE RESEARCH AIM

Develop an operational Machine Learning classification framework for daily rainfall patterns and extreme storms

using 10 years of synchronous DHM ground observations and NASA GPM IMERG V07 satellite data to empower municipal disaster resilience.

SPECIFIC RESEARCH OBJECTIVES

01 **Spatial Microclimatic Patterns & Topographic Regimes:**
Delineate microclimatic rainfall regimes across the watershed using Principal Component Analysis (PCA) and K-Means clustering, quantifying elevation lapse rate dependencies (+62.9 mm/100m).

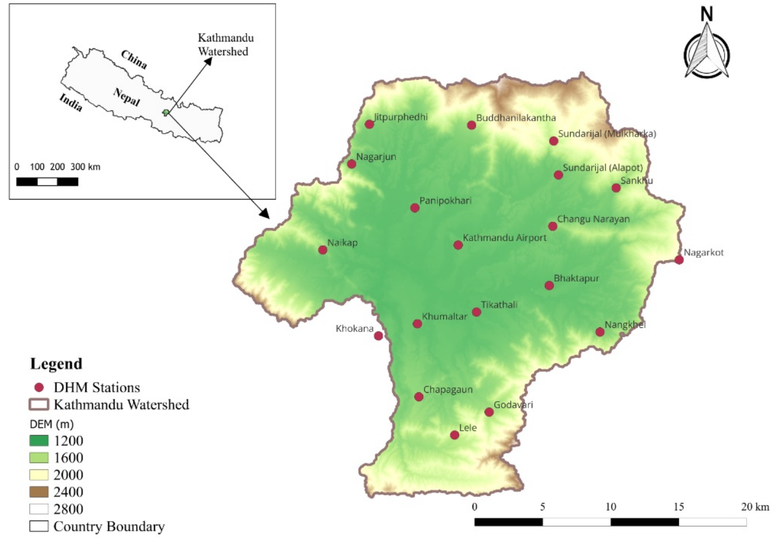
02 **Supervised ML Classification Under Extreme Class Imbalance:**
Develop, optimize, and benchmark Random Forest, XGBoost, and Support Vector Machines (Linear & RBF) across five intensity categories under severe class imbalance, interpreted via SHAP.

03 **Decadal Non-Parametric Climate Trend Detection (2016–2025):**
Assess decadal changes in annual totals, seasonal concentration, rainy-day frequency, and extreme events using Mann–Kendall and Sen's slope methods under rigorous statistical significance.

DECISION-SUPPORT IMPACT

- Multi-Agency Utility: Directly serves Department of Hydrology & Meteorology (DHM), National Disaster Risk Reduction Authority, and Kathmandu Metropolitan City flood response units via an operational interactive Streamlit decision dashboard.

STUDY DOMAIN: KATHMANDU WATERSHED & DHM NETWORK



GEOGRAPHICAL & OROGRAPHIC CHARACTERISTICS

- **Watershed Extent:** Latitudes 27.55°N–27.85°N, Longitudes 85.20°E–85.55°E; Drainage Area: ~650 km².
- **Elevation Profile:** 1,280 masl (Valley floor plain) to 2,147 masl (Nagarkot eastern ridge crest).
- **Network Density:** 19 screened DHM stations: Kathmandu (10), Lalitpur (6), Bhaktapur (3).
- **Rainfall Contrast:** Annual mean ranges from 1,000.5 mm (Tikathali) to 2,341.0 mm (Sundarijal Mulkharka).

RESEARCH CHALLENGES, IDENTIFIED GAPS & PROPOSED METHODOLOGY

Addressing Satellite Uncertainty, Severe Class Imbalance, and Orographic Microclimates

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01 CHALLENGE 1: SATELLITE UNCERTAINTY

- **Overestimation Bias:**
  Systematic positive bias of +2.78 mm/day (+46.12%) over Himalayan ridges.
- **High False Alarm Rate:**
  False Alarm Ratio (FAR) = 46.12%; satellite flags light rain when gauges are dry.
- **Ridge Cloudburst Error:**
  Coarse 0.1° (~11 km) gridded pixels miss localized high-intensity cloudburst cells.

02 CHALLENGE 2: SEVERE CLASS IMBALANCE

- **Light Rain Dominance:**
  Class 0 (<10 mm/d) represents 86.11% of all records (59,768 station-days).
- **Extreme Event Rarity:**
  Very Heavy (0.10%) and Extreme (0.013%) combined comprise only 0.11% (75 days).
- **Standard Model Failure:**
  Default loss functions maximize majority class accuracy, completely missing hazard events.

03 CHALLENGE 3: SPATIO-TEMPORAL COMPLEXITY

- **Steep Orographic Relief:**
  Elevation gradient from 1,280 to 2,150+ masl drives intense localized microclimates.
- **Monsoon Concentration:**
  >80% of total annual precipitation is concentrated within 120 days (June–September).
- **Shifting Decadal Baseline:**
  Non-parametric trend detection needed to decouple climate shifts from seasonal noise.

GAP 1: LACK OF POINT-TO-GRID CALIBRATION

Prior Literature Limitation:
Sharma et al. (2020) evaluated IMERG across Nepal but aggregated regionally, smoothing localized valley-ridge contrasts.

OUR NOVEL ADVANCEMENT →

Integrates station elevation and coordinates to calibrate satellite bias at specific gauge locations across steep terrain.

GAP 2: ISOLATED TRENDS WITHOUT PREDICTION

Prior Literature Limitation:
Khanal et al. (2024) analyzed decadal trends as retrospective statistics without connecting them to predictive classification.

OUR NOVEL ADVANCEMENT →

Unifies historical non-parametric trend modeling (Mann–Kendall/Sen) with multi-class forward-looking storm forecasting.

GAP 3: BLACK-BOX ML LACKING EXPLAINABILITY

Prior Literature Limitation:
Lei et al. (2025) benchmarked ML downscaling but treated models as black boxes without hydrological validation.

OUR NOVEL ADVANCEMENT →

Couples unsupervised spatial clustering (PCA/K-Means) with SHAP attribution to explain antecedent rainfall memory.

Problem Statement Conceptual Architecture:

KEY CHALLENGES IN RAINFALL ANALYSIS – KATHMANDU WATERSHED

Research problem and motivation for satellite comparison and machine learning

1. SPATIAL AND SEASONAL VARIABILITY
- Kathmandu watershed area: 587 km²
- Elevation ranges from about 1,200 m to over 2,800 m
- More than 80% of annual rainfall occurs during June–September
- Rainfall varies across locations and seasons

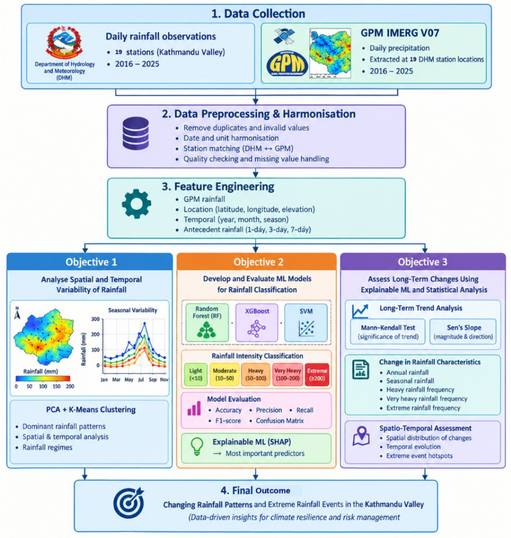
2. RAINFALL DATA LIMITATIONS
- DHM observations come from 19 stations
- GPM IMERG V07 provides wider spatial coverage
- Satellite estimates have uncertainty
- False Alarm Ratio: 46.12%
- Mean bias: +2.78 mm/day

3. RARE HEAVY-RAINFALL EVENTS
- Study period: 2016–2025
- Total observations: 69,407 station-days
- Light rainfall: 86.11%
- Very heavy and extreme rainfall: 0.11%
- Class imbalance can make rare events difficult for machine-learning models to identify

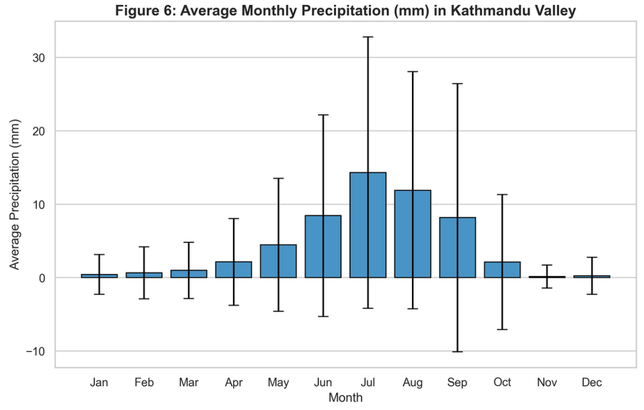
RESEARCH RESPONSE

DHM rainfall observations + GPM satellite estimates → Data preparation and comparison → Machine-learning rainfall classification → Identification of rare heavy-rainfall events

Integrated 7-Stage Machine Learning Computational Pipeline:



Monsoon Seasonality (2016–2025):



DATASET & METHODOLOGY SPECIFICATIONS

DATASET ENGINEERING, MICROCLIMATIC CLUSTERING & ML ARCHITECTURE

Spatio-Temporal Feature Structuring, 5-Tier Intensity Partitioning, and Unsupervised Topographic Regimes

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1. DATASET SCREENING & 5 INTENSITY TIERS

TERRESTRIAL & SATELLITE HARMONIZATION

- 21 Stations Screened: Tarebhir (100% missing) & Thankot (92.5%) dropped.
- 19 Valid Gauges Retained: 3,694 missing observations linearly imputed.
- NASA GPM IMERG V07: Synchronously matched at station coordinates (2016–2025).

OPERATIONAL RAINFALL INTENSITY TIERS & SAMPLE DISTRIBUTION

Class / Category	Threshold	Samples (N)	Share (%) & Proportion	Hazard Level
Class 0: Light	< 10 mm/d	59,768	**86.11%**	Routine
Class 1: Moderate	10–50 mm/d	8,820	**12.71%**	Advisory
Class 2: Heavy	50–100 mm/d	744	**1.07%**	Watch
Class 3: Very Heavy	100–200 mm/d	66	**0.10%**	Warning
Class 4: Extreme	≥ 200 mm/d	9	**0.013%**	Emergency

CHRONOLOGICAL DATASET PARTITIONING (LEAK-FREE)

Training Set (70.0%): **2016–2022 (7 Years) | N = 48,583 samples**

Full annual cycles establishing base parameters without future awareness.

Validation Set (20.0%): **2023–2024 (2 Years) | N = 13,889 samples**

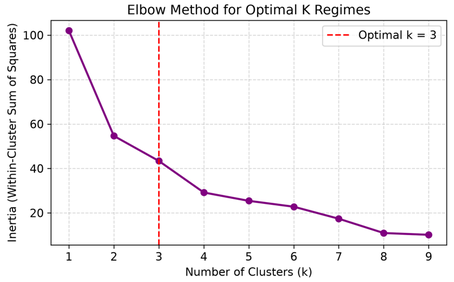
Hyperparameter optimization, probability threshold calibration & regularization.

Independent Test (10.0%): **2025 Holdout (1 Year) | N = 6,935 samples**

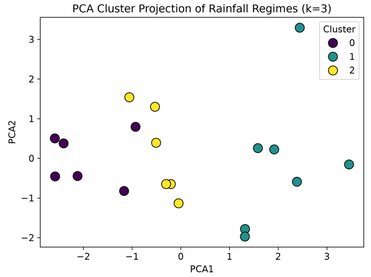
Strict forward temporal generalization benchmark simulating real deployment.

SPATIAL VALIDATION: Leave-One-District-Out (LODO) across KTM, LLT, BKT

2. UNSUPERVISED MICROCLIMATIC CLUSTERING



Elbow & Silhouette Score: Confirms K=3 as optimal regime partition (Silhouette = 0.584)



PCA Cluster Space: Visualizes clear geometric separation driven by elevation gradients

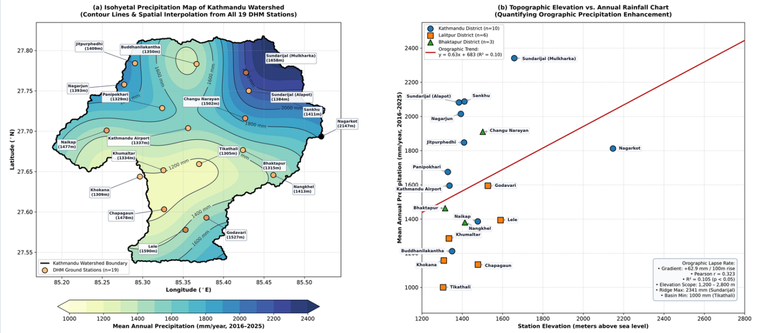
THREE DELINEATED MICROCLIMATIC RAINFALL REGIMES

Regime 0 (Valley Floor): 1,300–1,400 masl (10 stations: Airport, Khumaltar); 1,350–1,500 mm/year.

Regime 1 (Mid-Slopes): 1,400–1,600 masl (6 stations: Godavari, Lele, Chapagaun); 1,600–1,850 mm/year.

Regime 2 (Ridge-Tops): >1,600 masl (3 stations: Nagarkot, Sundarijal Mulkharka); >2,000 mm/year.

3. ISOHYET LAPSE RATE & MODEL SUITE



Isohyetal Map & Lapse Rate: Physical increase of +62.9 mm rain per 100 m elevation rise

EVALUATED SUPERVISED MACHINE LEARNING SUITE

Random Forest (RF): **Ensemble of 15 trees, max depth 3, balanced class weights.**

High variance reduction, resilient to feature noise, preserves rare storm classes.

XGBoost Classifier: **50 boosted trees, max depth 3, lr 0.01, multi:softprob.**

Regularized objective function minimizes over-fitting on minority storm classes.

Linear SVM (Scaled): **Linear Kernel Support Vector Classifier, C = 0.1, balanced.**

Maximizes geometric margin between classes in standardized feature space.

RBF SVM (Scaled): **Radial Basis Kernel Support Vector Classifier, C = 1.0, gamma = scale.**

Projects inputs into non-linear Hilbert space to form complex decision boundaries.

EXPERIMENTAL BENCHMARK & EVALUATION

EXPERIMENTAL RESULTS, MODEL BENCHMARKING & EXPLAINABLE AI (XAI)

Comparative Evaluation on Independent 2025 Test Set (N = 6,935) & SHAP Hydrological Attribution

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INDEPENDENT 2025 TEST SET BENCHMARK RESULTS (N = 6,935 STATION-DAYS, 19 DHM STATIONS)

Evaluated Classifier	Accuracy	Balanced Acc.	Weighted Prec.	Weighted Rec.	Weighted F1	Macro F1	Macro ROC-AUC
Random Forest	69.46%	**59.80% (Best)**	87.20%	69.46%	75.51%	0.3446	0.87
XGBoost Classifier	70.48%	45.00%	86.53%	70.48%	76.37%	0.3310	**0.88 (Best)**
Linear SVM (Scaled)	**74.72% (Best)**	59.54%	85.88%	**74.72% (Best)**	**78.92% (Best)**	**0.3682 (Best)**	0.84
RBF SVM (Scaled)	70.22%	42.62%	87.19%	70.22%	76.26%	0.3226	0.80

OPERATIONAL TRADE-OFFS & SELECTION LOGIC

Linear SVM (Operational Winner):

Highest accuracy (74.72%) and weighted F1 (78.92%) with instant sub-millisecond inference, optimal for routine municipal daily forecasts.

Random Forest (Life-Safety Winner):

Highest balanced accuracy (59.80%) and captured 80.00% (8/10) of Class 3 Very Heavy Storms, critical for flash flood mitigation.

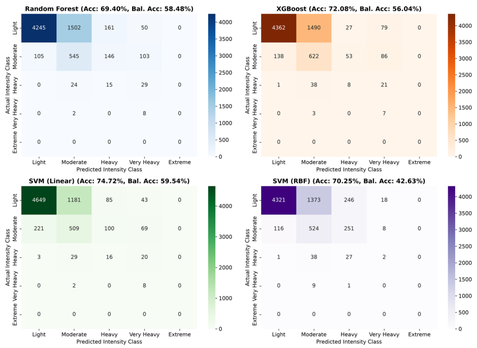
XGBoost (Discriminative Power):

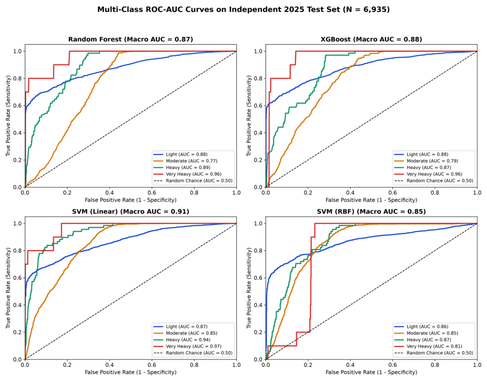
Delivered highest Macro ROC-AUC (0.88) and best Heavy Rain recall (42.65%), providing smooth calibrated risk probabilities.

Key Deployment Synthesis:

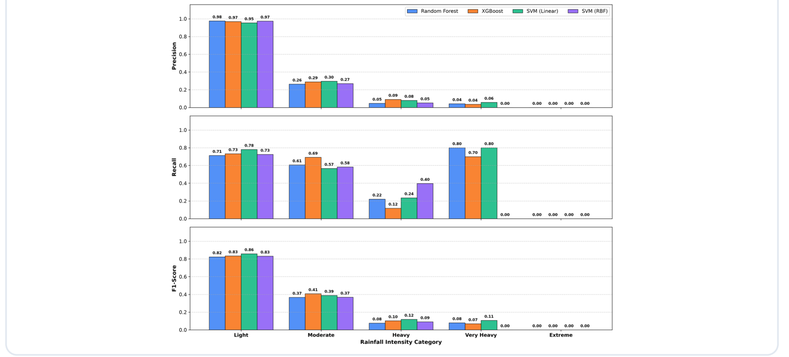
No single algorithm dominates all metrics; an operational ensemble or threshold-switched deployment balances precision with extreme event recall.

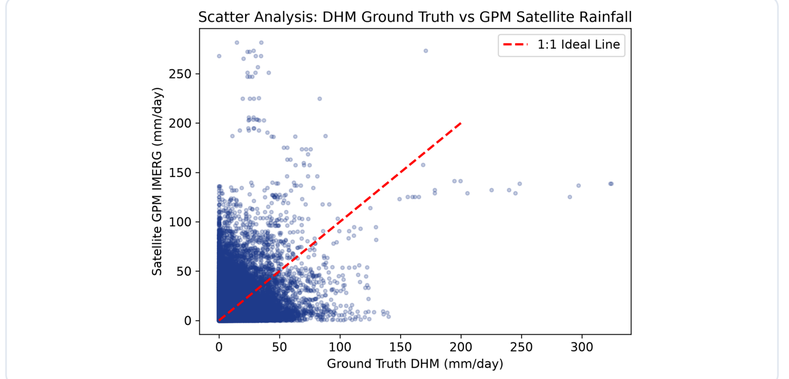
Multi-Model Evaluation Diagnostics (2025 Test):



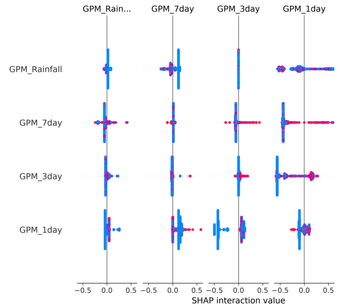


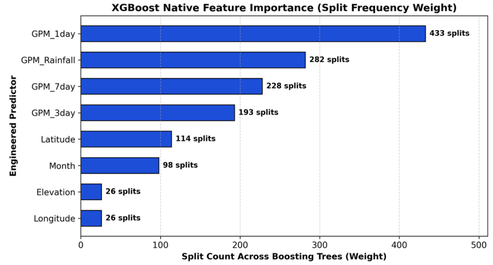
Per-Class Accuracy & Satellite Baseline Validation:





Explainable AI (XAI) & Antecedent Rainfall Memory:





PHYSICAL HYDROLOGICAL INSIGHTS FROM SHAP

- **Dominant Memory:**
  1-Day Lag (P_t-1) accounts for 74.47% relative feature gain.
- **Multi-Day Persistence:**
  3-Day Lag (P_t-3) provides 9.47%; concurrent satellite = 6.30%.
- **Physical Catchment Priming:**
  Prior rain saturates the mountain basin, primes soil moisture and humidity, triggering catastrophic high-velocity runoff.

CLIMATOLOGICAL TRENDS & OPERATIONAL DEPLOYMENT

DECADAL CLIMATE TRENDS, OPERATIONAL DASHBOARD & FUTURE WORK

Mann–Kendall Trend Verification (2016–2025), Streamlit Decision Support, and Scientific Conclusions

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DECADAL CLIMATE TREND DETECTION ACROSS KATHMANDU WATERSHED (2016–2025)

Climatological Metric	Kendall's Tau (τ)	p-Value	Sen's Slope (Q)	Statistical Trend Result	Climatological Significance
Annual Mean Rainfall	**+0.5111**	**0.0491***	**+0.2012 mm/year**	**Statistically Increasing**	**Significant at 95% Conf (p < 0.05)**
Maximum Daily Rainfall	+0.3556	0.1780	+2.4800 mm/year	Positive Upward Slope	Intensifying Peaks (Not Stat. Sig.)
Rainy Days (> 1 mm/day)	+0.3778	0.1524	+40.1667 days/year	Positive Upward Slope	Extended Wet Spells (Not Stat. Sig.)
Extreme Storms (> 100 mm/day)	+0.3778	0.1491	+0.8889 events/year	Positive Upward Slope	Frequent Cloudbursts (Not Stat. Sig.)

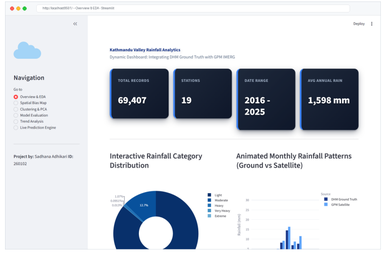
SCIENTIFIC RIGOR & ATTRIBUTION NOTICE

ACADEMIC HONESTY & CLIMATOLOGICAL BOUNDARY

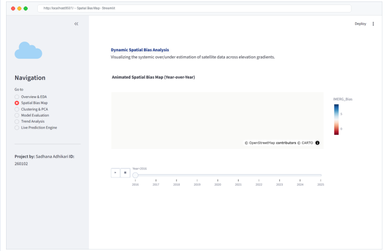
- **Statistical Significance:**
  Only Annual Mean Rainfall reached statistical significance (p = 0.0491 < 0.05). Maximum daily peaks and extreme events showed strong positive slopes (+2.48 mm/yr, +0.89 events/yr) but remained p > 0.05.
- **Attribution Boundary:**
  A 10-year observational series (2016–2025) reliably documents localized decadal climate variability and current hazard risk. Formal anthropogenic attribution requires multi-decadal records (≥30 years) and global coupled climate simulations.

OPERATIONAL RESEARCH DASHBOARD: INTERACTIVE DECISION-SUPPORT APPLICATION (dashboard.py)

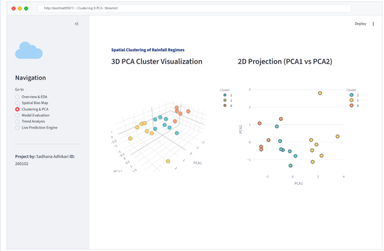
LOCALHOST DEPLOYED @ STREAMLIT ENGINE



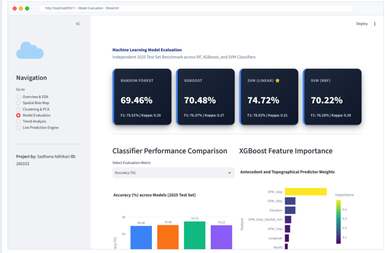
1. Overview & EDA
3D animated metric cards



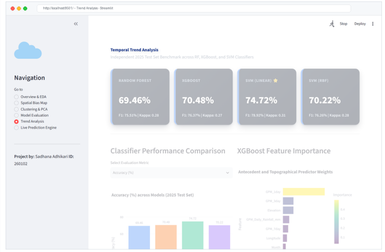
2. Spatial Bias
Geospatial error heatmap



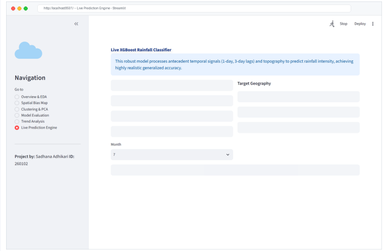
3. PCA Regimes
Interactive 3D scatter



4. Model Benchmarks
Confusion matrix & ROC



5. Decadal Trends
Time series & OLS slider



6. Live Inference
Real-time rainfall slider

KEY RESEARCH CONTRIBUTIONS & SCIENTIFIC FINDINGS

Integrated 10-Year Multi-Source Dataset:	Harmonized 69,407 station-days from 19 DHM gauges with NASA GPM IMERG V07 satellite.
Satellite Bias Profiling:	Proven that raw satellite systematically overestimates light rain (MB = +2.78 mm/d, FAR = 46.12%).
Topographic Microclimates:	Delineated 3 elevation rainfall regimes via PCA & K-Means (Silhouette = 0.584, lapse rate = +62.9 mm/100m).
Machine Learning Under Imbalance:	Linear SVM achieved 74.72% accuracy; Random Forest captured 80.00% of very heavy storms.
Hydrologic Explainability:	SHAP attribution revealed prior 1–3 day rainfall memory accounts for 83.94% of total predictive gain.
Operational Software Delivery:	Built an interactive Streamlit decision-support web dashboard for real-time hazard evaluation.

FUTURE HORIZONS & STRATEGIC ROADMAP

- **High-Resolution Radar Assimilation:**
  Integrate X-band dual-polarization weather radar for micro-scale convective tracking.
- **Deep Recurrent Architectures:**
  Evaluate LSTM and Temporal Convolutional Networks (TCN) for sub-daily hourly flash flood forecasts.
- **High-Altitude Ridge Monitoring:**
  Deploy solar-powered IoT tipping-bucket rain gauges across unmonitored ridgelines (>2,200 masl).
- **Municipal Early Warning Integration:**
  Connect ML model inference API directly into Kathmandu Metropolitan disaster dispatch.

CLOSING RESEARCH VISION

"Towards Explainable, Data-Driven and Spatially Aware Rainfall Intelligence for Climate-Resilient Kathmandu."